

CHAPTER-3

SHEET METAL WORK

3.1 Introduction

Sheet metal work is generally regarded as the working of metal from 16 gauge down to 30 gauge with hand tools and simple machines into various forms by cutting forming into shape and joining. It has its own significance as a useful trade in engineering works and also for our day to day requirements.

The sheet metal shop is very important for every engineering concern. It deals with the working of metal sheets. It requires a through knowledge of the projective geometry particularly the development of surfaces, because the laying out pattern and cutting of metal sheets to correct sizes and shapes entirely depends upon the knowledge of the workman. The various operations performed in a sheet metal shop are cutting, shearing bending etc. Allowance should be given in the drawing stage for folding and bending.

Common examples of sheet metal work are to prepare boxes, hoppers, cannisters, covers, funnels, cylinders, ducts, prisms, cones, pyramids, etc. from a flat sheet metal.

3.2 Metals use in sheet metal work

A variety of metal used in a sheet metal shop such as

Black Iron

The cheapest sheet metal is black iron, which is sheet iron rolled to the desired thickness, annealed, by placing in a furnace until red hot, and then set aside to cool gradually. It has a bluish-black appearance and is often referred to this metal is limited to articles that are to be painted or enameled such as tanks, pans, stove pipes, etc.

GALVANISED IRON

Zinc-coated iron is known as "galvanized iron". This soft steel sheet is popularly known as GI sheet. The zinc coating resists rust, improves the appearance of the metal and permits it to be soldered with greater ease ; but welding is not so easy as zinc gives toxic fumes and residues. Because it is coated with zinc, galvanized sheet iron withstands contact with water and exposure to weather. Articles such as pans, buckets, furnaces, heating ducts, cabinets, gutters etc. are made mainly from GI sheets.

STAINLESS STEEL

This is alloy of steel with nickel, chromium, and traces of other metals. It has good corrosive resistance and can be welded easily. The cost of stainless steel is very high. Stainless steel used in the sheet metal shop can be worked as galvanized iron sheets, but is tougher than GI sheets. This is used in canneries, dairies, food processing and chemical plants, kitchenwares, etc.

COPPER

Copper sheets are available either as cold-rolled or hot-rolled sheets. Cold-rolled sheets being worked easily are commonly used in sheet metal shop, and are resistant to corrosion. They have a better appearance than other metals. Copper being a costly metal, cost of copper sheets is higher in comparison of GI sheets. Gutters, expansion joints, roof flashing and hoods are some of the common examples of copper sheet.

ALUMINIUM

Aluminium cannot be used in pure form, but is used with a very small amount of copper, silicon, manganese and iron. It is highly resistant to corrosion and abrasion, whitish in colour and light in weight. It is now widely used in the manufacture of a number of articles such as household appliances, refrigerator trays, lighting fixtures,

windows, in the construction of airplanes, in the fitting and fixtures use in doors, window and building requirements, and in many electrical and transport industries.

TIN PLATE

Tin plate is sheet iron coated with the tin to protect it against rust. This is used for nearly all soldered work, as it is the easiest metal to join by soldering. The size and thickness of tin plates are denoted by special marks, not by gauge numbers, and can be very confusing to the uninitiated. If larger or heavier sheets of tinned iron are required, the material uses is known as "tinned steel", and this may be obtained in all sizes and gauges in which sheet iron is obtainable.

However, this metal has a very bright silvery appearance and is used principally in the making of roofs, food containers, dairy equipments, furnace fitting, cans and pans etc.

LEAD

Lead sheets are used in highly corrosive acid tanks. Lead is very soft and heavy. Lead sheet can be worked by hand without the use of any mechanical device.

3.3. HAND TOOLS

There is a fairly large number of hand tools used by sheet metal workers. The tools most commonly used are

1. Trammels :

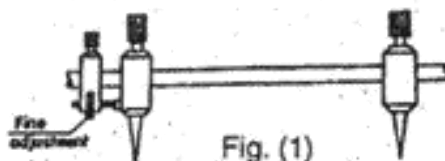


Fig. (1)

Sheet metal layout requires marking of arcs and circles. This may be done using the trammels as shown in fig. 1. The length of the beam decides the maximum size of the arc that can be scribed.

2. Wire Gauge

The wire gauge as shown in Fig (2), is used to check the diameter of wires or thickness of sheet metal. The commonly used wire gauge is imperial standard wire gauge known as SWG. It has slots to indicate different width sizes. The most common wire gauge used in sheet metal has 21 slots with gauges varying from 4 to 24. The following table shows the imperial standard wire gauge sizes.



Fig. 2

SWG No.	Size in mm	SWG No.	Size in mm	SWG No.	Size in mm
4	6.0	11	3.0	18	1.25
5	5.5	12	2.7	19	1.05
6	4.8	13	2.4	20	0.95
7	4.5	14	2.1	21	0.85
8	4.0	15	1.9	22	0.72
9	3.6	16	1.65	23	0.62
10	3.3	17	1.45	24	0.58

3. Snips



Fig. 3(a)

A snip, also called a hand shear is used 1) Like A PAIR OF SCISSORS To cut thin soft metal. It should be used only to cut 20 gauge or thinner metal. There are several types of snips available for making straight or circular cuts, the most common being straight snips and curved snips.

The straight snip as shown in Fig 3(a) have straight blades for straight line cut-

ting while curved or bent snips have curved blades for making circular cuts.



Fig. 3 (b)

The curved snips, as shown in fig. 3 (b) have curved blades for making circular cuts. They are available for either right hand or left hand cuts.

4. Hammers

Light weight hammer and mallets used in sheet metal work. Ball peen hammer has a cylindrical, slightly curved face and a ball head. It is a general purpose hammer, used mostly for riveting in sheet metal work. The cross peen hammer has a tapered peen and is perpendicular to the handle. Because of this, it can reach backward corners.

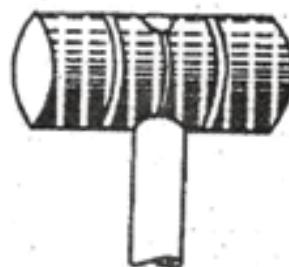


Fig. 4

Straight peen hammer has the peen end similar to the cross peen, but it is positioned parallel to the handle which can be used conveniently for certain operations of folding.

The mallets may be made from hide, fibre or wood. The best size of mallet is 5 cm diameter. These may be obtained in various shapes to suit specific work as shown in fig. (4) mallet is used for bending and folding work. It is light in weight, covers more area and does not produce dent on sheet metal.

5. Stakes

Stakes are the sheet metal worker's anvils used for bending, seaming or forming, using a hammer or mallet. They actually work as supporting tools as well as forming tools. They also help in bending operation. They are made in different shapes and

sizes to suit the requirements of the work. The beakhorn stake is used for riveting, forming around and square surfaces, bending straight edges, and making corners. The hatchet stake is used to make straight, sharp bends and for folding and bending edges.

5) Stakes :-

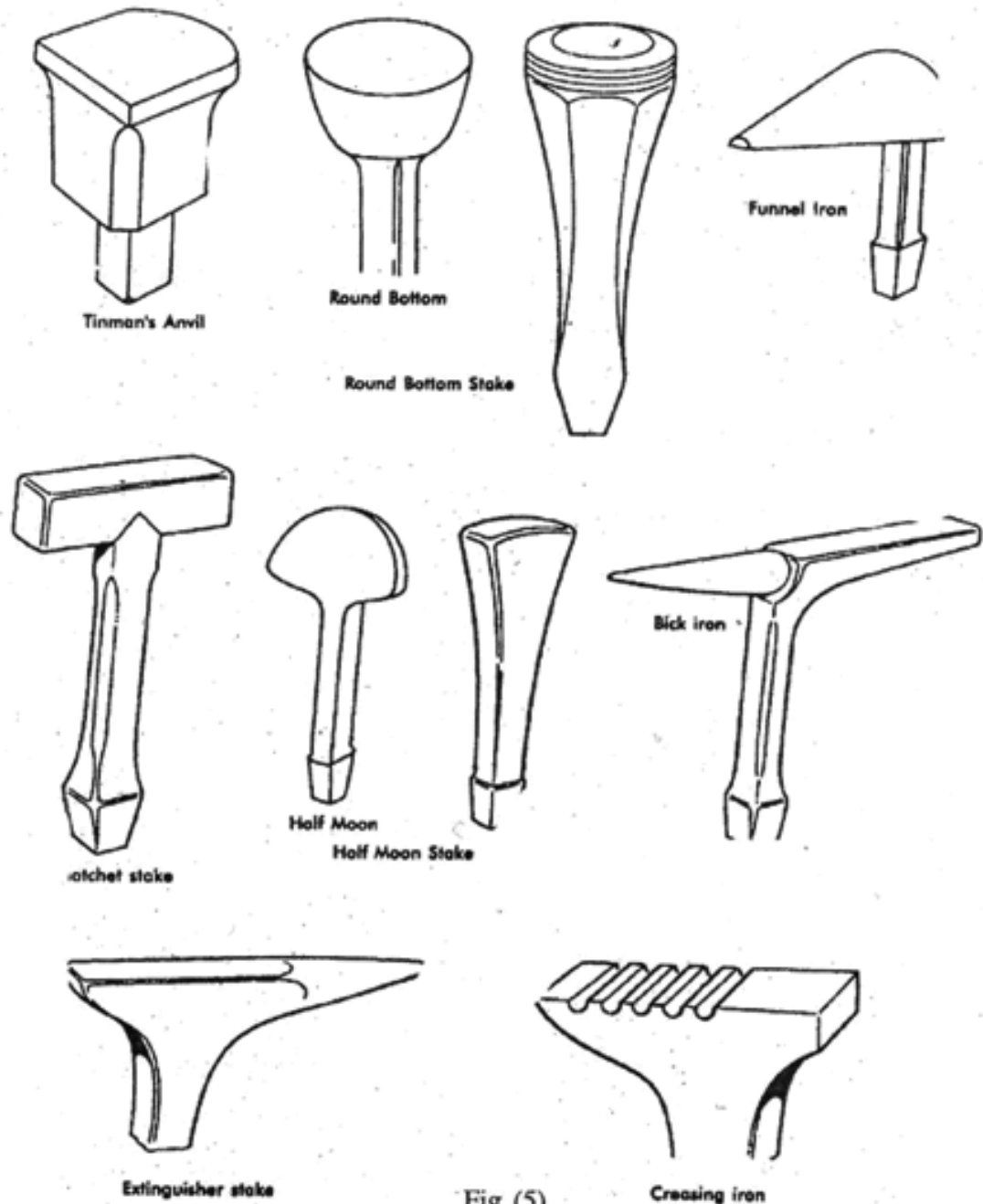


Fig. (5)

3.4. Steel metal joints

Various types of joints are used in sheet metal work, to suit the varying requirements. Some commonly used sheet metal joints and folded edges are shown in fig. (6).

These are self secured joints formed by joining together. Two pieces of sheet metal are used using the metal itself to form the joint. These joints are to be used on sheets of less than 1.6 mm thickness. Usually in workshop the commonly used sheet metal are soldering by using adhesives, riveting or mechanical joint.

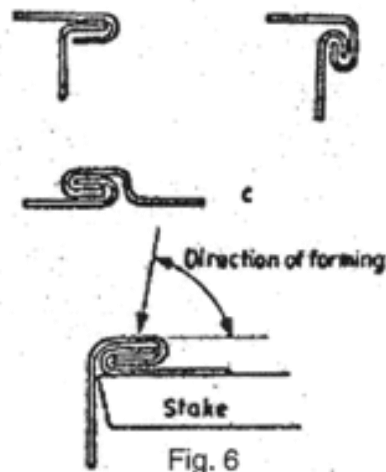


Fig. 6

3.5 SOLDERING

Soldering is one method of joining two pieces of metal with an alloy that melts at a lower temperature than the metals to be joined. For a good job, the metals to be joined must be free from dirt, grease and oxide. Solder is made of tin and lead, usually in equal proportions. It comes in the form of wire.

Capillary action between the solder and the base metal will not take place unless the base metal is clean. A flux is used with the solder to remove oxides and thus permit good solder wetting. An acid flux is used for black iron and galvanized iron. A rosin flux is used for tin plate, copper and electric wire.

Soldering is advantageous as similar and dissimilar metals can be joined by this process.

3.5.1 Soldering Iron

Soldering requires a source of heat. A common method of transmitting heat to the metal surfaces is by using a soldering iron. Figure (7) shows the soldering iron

used for the purpose. The working end of this tools is made of copper, which is a good conductor of

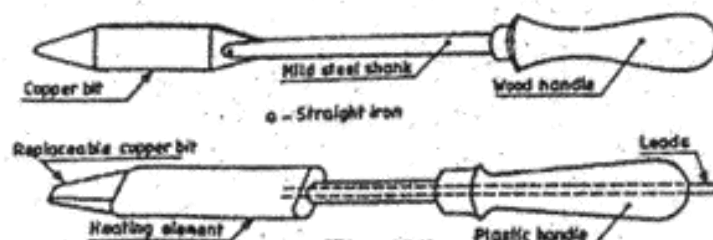


Fig. (11)

3.5.2 Method of soldering

The following are the stages involved in soldering work :

1. Clean the surfaces to be soldered.
2. Keep the surfaces to be joined, close together.
3. Apply a thin layer of flux with a brush.
4. Heat the soldering copper to proper temperature.
5. Tack the seam by applying solder at several points.
6. Begin at one end and move the copper bit slowly, adding solder as needed.
7. Allow the joint to cool.

3.5.3 Reasons for using Soldering Flux

1. To clean the surface of the base metal during heating.
2. To eliminate impurities present in metals.
3. To break down the surface tension of the metal and enable the solder to flow.
4. To prevent the formation of fresh oxides, by forming a protective layer on the solder.

3.5.4 Uses of Soldering

Soldering is used to join the following :-

1. Electrical components in television, radio, transistor and tape recorders
2. Electronic components like printed circuit boards
3. Automobile parts like radiators
4. Sheet metal works
5. Utensil repairs

3.6. BRAZING

Brazing is a process of joining two metal pieces by the addition of non ferrous filler metal. By definition, brazing employs a filler metal with melting temperature above 150°C. Copper and copper alloys, silver and silver alloys and aluminium alloys are the most commonly used filler metals for brazing.

The surfaces to be joined must be cleaned first, by removing all grease and oxide. After cleaning, a flux is applied to all surface where the filler metal is a flow. Fluxes are used to prevent oxidation of the base metal and filler metal during brazing and promote the free flowing of the filler metal by capillary attraction. The commonly used flux for brazing are borax, boric acid, borates, fluorides, and chlorides.

In brazing the parts to be joined and the filler metal are brought too temperature which is below the melting point of the parts to be joined, but above the melting point of the brazing material. The filler metal flows throughout the joint by capillary attraction and allowed to solidify. Since fluxes are corrosive in nature, they should be removed after brazing. Brazing is used for the fastening of pipe fittings, tanks, radiators, heat exchangers, carbide tips on tool holders, etc.

Braze welding is similar to ordinary brazing except that the filler metal is not distributed by capillary attraction. In this process, filler metal is melted and deposited at the point where the weld is to be made.

3.6.1 Advantage of Brazing

1. Dissimilar metals and parts having thin sections can be joined easily.
2. Brazing may avoid the metallurgical damage to the metal.
3. Strong join can be obtained by brazing.
4. Properly brazed joints are pressure tight.
5. It is an economical and quick process.
6. Less heating is required than for welding.

3.6.2 Disadvantages of Brazing

1. Large areas cannot be brazed easily due to the possibility of lack of uniform heating of internal surfaces.
2. Joints have poor strength compared to welded joints.
3. Require tightly mating parts.
4. Brazing fluxes may produce toxic fumes.

3.6.3 Uses of Brazing

1. To Join carbide tips with steel shanks.
2. To fasten pipe fittings, tanks, heat exchanges and electrical repair work.
3. To join cycle parts and steam turbine components.

3.6.4 Difference between Brazing soldering

Brazing	Soldering
1. Melting point of the filler material is above 450°C	Melting point of the filler materials is below 450°C.
2. Dissimilar metals can be joined easily.	Only similar metals can be joined
3. Good surface finish.	Does not yield a good surface finish
4. Stronger joints.	Less stronger joints
5. The strength of the brazed joint is a function of the attraction forces between the molecules of the brazing material.	The solder acts as a metal solvent by melting small amounts of the base metal, and these form a chemical bond with one of the solder constituents. Thus
the strength of the joint is function of the	alloy formed.